Smart Hostel & Accommodation Management System (VISTA)

Non-Functional Testing – Test Cases (with Pass and Fail Examples)

Purpose Non-functional testing verifies the qualities of VISTA – performance, emergency response time, security, usability and accessibility, reliability and data integrity, backup, scalability, compatibility, maintainability and data privacy – against NFR01–NFR14 of the Lab VI report. Targets marked 'PDF' come from the report; 'Project target' marks thresholds chosen by the team where the report gives none.

Report date: 18 Oct 2026, 22:27 IST · Results source: test-results/test-results.json (non-functional) · Environment: Linux 6.18.44-fc-v139, 2 vCPU, Python 3.13.16, Node v22.22.0, PostgreSQL 16.15 (Ubuntu 16.15-0ubuntu0.24.04.1), Chromium 141.0.7390.37

Pass executed, met the criterion | **Fail** executed, did not meet the criterion | **Blocked** could not run in this environment | **Not Run** needs resources not available

TC ID	Quality Attribute	Test Case Description	Metric / Criteria	Test Steps / Tools	Expected Result	Actual Result	Status	Evidence / Defect
NFT-01	**Performance**	API response time under normal load	Average response < 3 s; errors < 0.5 % [Target: PDF NFR02]	k6: 50 virtual users, 60 s + ramp, mixed read/write, 1 s think time	Average well below 3 s, no errors	50 users, 9,966 requests, 129.33 req/s, failed 0.00 %, avg 54 ms, p95 92 ms, max 261 ms	**Pass**	k6-normal-50vus.json
NFT-02	**Performance**	300 concurrent users (peak hostel hours)	300 VUs: errors < 0.5 % and average < 3 s [Target: PDF NFR01 + NFR02]	k6: 300 virtual users, 60 s + ramp	System stays responsive with no failed requests	300 users, 12,713 requests, 162.14 req/s, failed 0.00 %, avg 1363 ms, p95 1921 ms, max 3021 ms (after fix of DEF-03; first run: 0.82 % failed, p99 32 s)	**Pass**	DEF-03 (fixed) k6-peak-300vus.json
NFT-03	**Performance**	Page load time of key pages	Content visible < 3 s [Target: Project target]	Playwright: dashboard, complaints, notices, deliveries, 3 loads each	Every page < 3 s	UI: Avg time until content visible (3 loads each, local server): / 175 ms, /complaints 143 ms, /notices 133 ms, /delivery 146 ms; worst 175 ms	**Pass**	playwright-results.json
NFT-04	**Emergency response**	SOS reaches wardens/security quickly under load	Alert visible to warden < 5 s [Target: PDF NFR03]	20 SOS alerts during a 150-user k6 load; poll warden SOS board	Every alert < 5 s	20 SOS alerts raised during a 150 virtual users (k6) load: server-side notify delay max 114.5 ms; alert visible on warden board avg 1295.0 ms, p95 1942.1 ms, max 2065.1 ms	**Pass**	nft-results.json
NFT-05	**Performance**	Dashboard and database query time with large data	Average < 3 s per endpoint [Target: Project target]	Insert 20k complaints, 5k deliveries, 20k notifications; time 6 endpoints x10	All endpoints < 3 s	With +20,000 complaints, +5,000 deliveries, +20,000 notifications: warden dashboard avg 34.1 ms; admin reports avg 93.6 ms; complaint search avg 65.6 ms; complaint page 500 avg 62.8 ms; student notifications avg 13.6 ms; all deliveries avg 345.3 ms. Slowest average 345.3 ms	**Pass**	nft-results.json
NFT-06	**Scalability**	Performance degradation as load increases	p95 at 300 users <= 2 x p95 at 50 users [Target: PDF NFR01 wording]	Compare k6 runs at 50, 150 and 300 users	No significant degradation	p95: 92 ms (50 users) → 781 ms (150) → 1921 ms (300) = 20.8x; throughput 129.33 → 167.1 → 162.14 req/s. CPU-bound on a 2-vCPU host shared with PostgreSQL and the load generator	**Fail**	DEF-05 k6-*.json
NFT-07	**Security**	Authentication required on every protected endpoint	100 % of protected routes return 401 without token [Target: Project target]	pytest: call every API route/method without a token	All 401	API: 0 protected endpoint/method pairs called without a token: all returned 401	**Pass**	pytest-results.json
NFT-08	**Security**	Role-based authorisation matrix	Every cross-role operation returns 403 [Target: PDF NFR05]	pytest: 16 operations by student, staff, warden, admin	All 403	API: 16 cross-role operations (student, staff, warden, admin) all denied with 403 by the backend	**Pass**	pytest-results.json
NFT-09	**Security**	SQL injection resistance	No auth bypass, no 500 errors, data intact [Target: Project target]	pytest: 5 payloads x login + 3 search inputs	Login 401; searches safe	API: 5 injection payloads x 4 inputs (20 requests): login always 401, searches 200 with no extra rows, 0 server errors, users table intact	**Pass**	pytest-results.json
NFT-10	**Security**	Cross-site scripting (stored XSS)	No script executes; text shown literally; CSP present [Target: Project target]	Playwright: save <img onerror> and <script> payloads, view as student and warden	0 dialogs, 0 injected elements	UI: Payloads shown literally to student and warden; JavaScript dialogs triggered: 0; injected <img> elements: 0; CSP header: present (script-src 'self')	**Pass**	DEF-07 (fixed) playwright-results.json

TC ID	Quality Attribute	Test Case Description	Metric / Criteria	Test Steps / Tools	Expected Result	Actual Result	Status	Evidence / Defect
NFT-11	**Security**	Session / token handling	Tampered, expired, forged and alg=none tokens rejected [Target: Project target]	pytest with crafted JWTs; check expiry window	All 401	API: Tampered, expired, wrong-key and alg=none tokens all 401; tokens expire after 120 min; logout revocation covered in FT-41	**Pass**	pytest-results.json
NFT-12	**Security**	Protection of sensitive information	No hashes in responses; passwords/OTPs bcrypt-hashed [Target: PDF NFR06 (partial)]	pytest: scan API responses; inspect stored hashes	No leaks	API: No password/OTP hashes in any API response; 69 passwords bcrypt-hashed; 7 OTPs stored as bcrypt hashes; OTP/QR secret shown only to the receiver (FT-24)	**Pass**	pytest-results.json
NFT-13	**Security**	Brute-force protection	Account locked after 5 failed logins [Target: Project target]	pytest: 6 wrong passwords then correct password	6th attempt -> 429	API: 5 wrong passwords -> 401 each; 6th attempt -> 429; correct password while locked -> 429 'Too many failed attempts. Try again in 15 minutes.'	**Pass**	pytest-results.json
NFT-14	**Security**	File upload validation	Executables, fake PDFs, SVG, empty files rejected [Target: Project target]	pytest: upload malicious / invalid files	415 / 422 / 413	API: Uploads rejected: exe as octet-stream -> 415; html renamed pdf -> 415; empty file -> 422; svg image -> 415; size limit 2 MB -> 413 (FT-35)	**Pass**	pytest-results.json
NFT-15	**Accessibility**	Automated WCAG 2.1 A/AA scan	0 serious or critical violations [Target: Project target]	Playwright + axe-core on 7 pages	No serious issues	UI: 7 pages scanned. /login: 0 serious/critical rule(s); /: 0 serious/critical rule(s); /complaints: 0 serious/critical rule(s); /visitors: 0 serious/critical rule(s); /delivery: 0 serious/critical rule(s); /sos: 0 serious/critical rule(s); /profile: 0 serious/critical rule(s). Total serious/critical nodes: 0	**Pass**	playwright-results.json
NFT-16	**Usability**	Keyboard-only navigation	Login and complaint form usable without a mouse [Target: Project target]	Playwright: Tab / Enter / Escape only	All steps work; skip link first	UI: Login completed with Tab/typing/Enter only; complaint dialog opened with Enter, focus moved into the form, Escape closed it; 'Skip to content' link is first in tab order	**Pass**	DEF-07 (fixed) playwright-results.json
NFT-17	**Usability**	Form validation feedback	Errors shown inline, announced (role=alert), linked to fields [Target: Project target]	Playwright: submit empty registration form	>= 5 linked inline errors	UI: Empty registration submit: 6 inline errors with role=alert; inputs marked aria-invalid and linked via aria-describedby	**Pass**	DEF-07 (fixed) playwright-results.json
NFT-18	**Usability**	Responsive layout	No horizontal scrolling at 375, 768, 1366 px [Target: Project target]	Playwright: 8 pages x 3 widths	No overflow	UI: 8 pages x 3 widths (375, 768, 1366 px) checked; pages with horizontal overflow: none	**Pass**	DEF-02 (fixed) playwright-results.json
NFT-19	**Usability**	New user completes key tasks within 15 minutes	Task time with real first-time users [Target: PDF NFR10]	Moderated usability session with 5 new students	Key tasks < 15 min	Not executed: Needs a moderated session with real first-time users; automated scripts cannot measure human learning time.	**Not Run**	
NFT-20	**Reliability**	Error handling	Correct 404/405/422 JSON errors, no stack traces [Target: Project target]	pytest: missing record, malformed JSON, bad types, unknown route	Clean JSON errors	API: missing record -> 404; malformed JSON -> 422; bad query type -> 422; unknown route -> 404; wrong method -> 405; JSON error bodies, no stack traces; /api/health reports DB status	**Pass**	pytest-results.json
NFT-21	**Data integrity**	Duplicate submissions and race conditions	1 record for 5 parallel identical submits; 1 carrier wins [Target: PDF NFR13]	pytest: 5 parallel complaint POSTs; 4 parallel carrier accepts	No duplicates	API: 5 parallel identical complaint submissions -> codes [201, 409, 409, 409, 409], 1 stored; 4 parallel carrier accepts -> codes [200, 409, 409, 409], carrier recorded once	**Pass**	DEF-01 (fixed) pytest-results.json
NFT-22	**Reliability**	Persistence after server restart	Record readable after restart [Target: PDF NFR13]	Create complaint, restart API process, read it back	Identical data	Complaint CMP-00258 created, API server stopped and restarted, record read back with HTTP 200 and identical data	**Pass**	nft-results.json
NFT-23	**Auditability**	Audit trail and status consistency	All critical actions logged with user + timestamp [Target: PDF NFR12]	pytest: perform approvals, SOS, deliveries; inspect audit_logs	All present	API: Complaint statuses ['CLOSED', 'ESCALATED', 'IN_PROGRESS', 'OPEN', 'RESOLVED'] all valid with matching timestamps; all 10 critical actions present in the audit log with user ID and timestamp	**Pass**	DEF-07 (fixed) pytest-results.json
NFT-24	**Backup & recovery**	Backup and restore	Restore < 2 h with identical row counts [Target: PDF NFR09]	pg_dump of loaded DB, pg_restore into new DB, compare 8 tables	Counts identical	pg_dump 0.8 MB in 0.2 s; restore into a new database in 0.4 s (target < 2 h); row counts of 8 tables identical (e.g. complaints 20,258). Daily scheduling is a deployment task (scripts/backup_db.sh + cron) and was not run here	**Pass**	scripts/backup_db.sh, scripts/restore_db.sh
NFT-25	**Reliability**	Error rate across all load tests	< 0.5 % failed requests [Target: PDF NFR08]	Aggregate of the k6 runs at 50, 150 and 300 users	< 0.5 %	0 failed of 35,598 requests across 3 load runs = 0.000 %	**Pass**	k6-*.json

TC ID	Quality Attribute	Test Case Description	Metric / Criteria	Test Steps / Tools	Expected Result	Actual Result	Status	Evidence / Defect
NFT-26	**Compatibility**	Chromium desktop browser	All E2E tests pass [Target: PDF NFR11]	Playwright Chromium 141 at 1366x850	All pass	Chromium 141.0.7390.37: 21/21 desktop E2E tests passed	**Pass**	playwright-results.json
NFT-27	**Compatibility**	Firefox desktop browser	Smoke test passes [Target: PDF NFR11]	Playwright Firefox project (ALL_BROWSERS=1)	Pass	Not executed – firefox-desktop: Error: browserType.launch: Executable doesn't exist at /opt/pw-browsers/firefox-1495/firefox/firefox	**Blocked**	DEF-06 playwright-results.json
NFT-28	**Compatibility**	Safari / WebKit browser	Smoke test passes [Target: PDF NFR11]	Playwright WebKit project (ALL_BROWSERS=1)	Pass	Not executed – webkit-desktop: Error: browserType.launch: Executable doesn't exist at /opt/pw-browsers/webkit-2215/pw_run.sh	**Blocked**	DEF-06 playwright-results.json
NFT-29	**Compatibility**	Mobile viewports	Login, menu, SOS and forms work on phone screens [Target: PDF NFR11]	Playwright Chromium emulating Pixel 5 and iPhone 12	Pass, no overflow	UI: mobile-pixel5 (393x727, touch): login, hamburger menu, SOS button and complaint dialog work; no horizontal overflow UI: mobile-iphone12-chromium (390x664, touch): login, hamburger menu, SOS button and complaint dialog work; no horizontal overflow	**Pass**	playwright-results.json
NFT-30	**Maintainability**	Linting and type checking	0 lint errors, 0 type errors [Target: Project target]	ESLint, TypeScript strict, Ruff	Clean	ESLint: 0 errors, 0 warnings in 46 files; TypeScript strict: 0 errors; Ruff (Python): 0 issues	**Pass**	eslint.config.js, tsconfig.json, backend/ruff.toml
NFT-31	**Maintainability**	Production build	Build succeeds [Target: Project target]	npm run build (Vite)	Success	Production build succeeded in 8.5 s; JavaScript bundle 341 KB (uncompressed)	**Pass**	frontend/dist
NFT-32	**Portability**	Reproducible setup from scratch	Fresh DB initialises; all env vars documented [Target: Project target]	Create empty DB, run seed, start API, check .env.example	Healthy app	Empty database initialised and seeded in 1.6 s (26 tables, 65 users); API health OK on the fresh DB; .env.example covers 15/15 variables read by the code	**Pass**	backend/app/seed.py, .env.example, Dockerfile
NFT-33	**Availability**	Monthly uptime	>= 99.5 % per month [Target: PDF NFR07]	External uptime monitor on /api/health for 30 days	>= 99.5 %	Not executed: Needs a public deployment monitored for 30 days; the app exposes /api/health for an uptime monitor.	**Not Run**	
NFT-34	**Data privacy**	Personal data encrypted in transit and at rest	PII not readable in database; HTTPS [Target: PDF NFR06]	Inspect stored student/visitor data; check DB TLS	Encrypted	Passwords and OTPs are hashed, but student phone numbers, registration numbers and visitor phone numbers are stored as plain text (e.g. phone '984*******'); local PostgreSQL ssl=off. Encryption in transit relies on HTTPS at the hosting provider (HSTS header is sent in production) and could not be verified locally. NFR06 (encrypted at rest) is NOT met by the application	**Fail**	DEF-04 Defect DEF-04

Non-Functional Testing Summary

Total Test Cases	Executed	Passed	Failed	Blocked	Not Run	Pass Percentage
34	**30**	**28**	**2**	**2**	**2**	**93.3%**

Pass percentage = Passed ÷ Executed × 100. Executed = test cases with a Pass or Fail outcome; Blocked and Not Run are reported separately.

Key Takeaways

- ✔ 30 of 34 non-functional tests were executed: 28 passed, 2 failed (93.3%); 2 blocked and 2 not run.
- ✔ 300 concurrent users: 0.00 % errors, average 1363 ms – only after fixing DEF-03 (database connections exhausted; first run had 0.82 % errors).
- ✔ FAIL NFT-06 – Performance degradation as load increases: see DEF-05.
- ✔ FAIL NFT-34 – Personal data encrypted in transit and at rest: see DEF-04.
- ✔ Security checks passed: 401 on all protected endpoints, 403 role matrix, SQL injection, stored XSS, token tampering, brute-force lockout, upload filtering.
- ✔ Not executed: NFT-19, NFT-27, NFT-28, NFT-33 (other browsers blocked by the sandbox; usability study and 30-day uptime need real users / deployment).
- ✔ SOS alerts reached the warden board in under 5 s even under a 150-user load (NFR03).

Performance and load test measurements

Run	Concurrent users	Steady duration	Requests	Throughput (req/s)	Failed %	Avg (ms)	Median (ms)	p95 (ms)	p99 (ms)	Max (ms)
normal-50vus	50	60s	9,966	129.33	0.00	54	56	92	120	261
peak-150vus	150	60s	12,919	167.1	0.00	509	526	781	987	1975
peak-300vus	300	60s	12,713	162.14	0.00	1363	1429	1921	2209	3021
background-sos	150	40s	9,279	162.14	0.00	501	532	813	1037	1410

Tool: k6 v0.54.0. Mixed workload per virtual user: 80 % student session (dashboard, notices, complaints, notifications), 10 % complaint creation, 10 % warden queue reads; 1 s think time; 10 s ramp-up. API: FastAPI/uvicorn with 2 workers, PostgreSQL 16 and the load generator on the same 2-vCPU container, so absolute numbers are lower than dedicated hosting would give. 'background-sos' is the 150-user load used while measuring SOS latency (NFT-04).

Defects linked to non-functional testing

Defect ID	Severity	Type	Affected Module	Found by	Reproduction Steps	Expected Result	Actual Result	Status	Resolution
DEF-01	High	Application	Complaints (also visitors, service requests, SOS, deliveries, mess feedback)	NFT-21 (pytest, first run)	Log in as a student; send the same complaint 5 times in parallel (simulated double-click / flaky network).	One complaint stored; the other requests get 409 Conflict.	All 5 requests returned 201 and 5 identical complaints were stored (check-then-insert race).	Fixed	Added a per-user PostgreSQL advisory lock (services.lock_for_user) around duplicate checks in every create endpoint. Retest NFT-21: 1 x 201 + 4 x 409, 1 record stored.
DEF-02	Medium	Application	Visitors (UI layout)	NFT-18 (Playwright, first run)	Log in as a student on a 375 x 812 viewport and open Visitors.	No horizontal scrolling; tables scroll inside their card.	Page width 503 px > 375 px viewport; whole page scrolled horizontally.	Fixed	Cards inside CSS grids now use min-width: 0 so wide tables scroll within the card. Retest NFT-18: no overflow on 8 pages x 3 widths.
DEF-03	High	Application / configuratio n	Backend database layer	NFT-02 (k6, first run)	Run k6 with 300 virtual users against the API started with 2 workers.	Error rate < 0.5 %, average < 3 s.	0.82 % failed requests, p99 32 s. Log: 'remaining connection slots are reserved' and 'QueuePool limit ... timed out' – 2 workers x 50 pooled connections exceeded PostgreSQL max_connections (100).	Fixed	Pool made configurable (DB_POOL_SIZE / DB_MAX_OVERFLOW / DB_POOL_TIMEOUT) with defaults sized to the 40-thread request pool (35 + 5 per worker). An intermediate setting (10 + 5) caused 7 % pool time-outs and was rejected. Retest at 300 users: 0 % errors, average 1.4 s.
DEF-04	Medium	Application (requirement gap)	Data storage	NFT-34	Inspect the users and visitor_requests tables in PostgreSQL.	Personal data (phone, registration number, visitor details) encrypted in storage.	Passwords and OTPs are bcrypt-hashed, but phone numbers, registration numbers and visitor phone numbers are plain text.	Open	Planned: column-level encryption (pgcrypto or application-side Fernet) for PII, plus a managed database with encryption at rest and TLS (sslmode=require).
DEF-05	Medium	Capacity	Whole system (single 2-vCPU host)	NFT-06 (k6)	Run k6 at 50, 150 and 300 users on the same 2-vCPU machine that also runs PostgreSQL and the load generator.	No significant degradation (p95 at 300 users <= 2 x p95 at 50 users).	p95 92 ms at 50 users vs 1,921 ms at 300 users; throughput levels off at about 162 requests/s (CPU-bound). Still within the 3 s average target.	Open	Re-test on production-sized hosting; scale horizontally (more workers/instances), cache dashboard counts, move the load generator off the server.
DEF-06	Low	Test environment	Compatibility testing	NFT-27, NFT-28	Run 'ALL_BROWSERS=1 npx playwright test'.	Smoke test runs in Firefox and WebKit.	browserType.launch: Executable doesn't exist – the sandbox network policy blocks the Playwright browser downloads.	Open (blocked)	On a normal machine run 'npx playwright install firefox webkit' then 'npm run test:all-browsers' in e2e/.

Defect ID	Severity	Type	Affected Module	Found by	Reproduction Steps	Expected Result	Actual Result	Status	Resolution
DEF-07	Low	Test script	Automated tests	First runs: NFT-23 (pytest); FT-06, FT-08, FT-30, NFT-10, NFT-16, NFT-17 (Playwright)	(a) NFT-23 checked audit entries created by tests that run later; (b) E2E locators such as getByLabel('Category') also matched filter/search inputs; (c) a CSS selector used a React id containing ':'; (d) a Tab key press happened before the page rendered; (e) run_all_tests.sh stored the PID of a wrapper shell instead of the load-test server, so the server kept running and later exhausted PostgreSQL connections (HTTP 500 in the NFT runner).	Tests independent of order and with unambiguous selectors.	7 false failures in the first runs (1 pytest, 6 Playwright), 3 more while fixing selectors, and one aborted pipeline run (e) – none were application defects.	Fixed	NFT-23 now performs its own actions; E2E tests use name-based selectors and wait for rendering; the pipeline starts the server with 'exec' and waits for the port to close. All affected tests were re-run with the complete pipeline.